

Exercise – Understanding the confidence interval (CI) using simulation

1. Aim of the exercise

The rationale behind a CI is based on the concept of repeated sampling from the population. However, in practical scenarios, we cannot repeatedly sample from the population, which makes the concept abstract and challenging to teach or learn. Consequently, students struggle to interpret the CI. Using simulation, we can mimic the process of repeated sampling from the population and demonstrate what a CI represents.

2. Theory

Assume that the model of interest is a standard linear regression model:

$$y_i = \beta x_i + u_i$$

If the error of the model is assumed to follow a normal distribution, the OLS estimator $\hat{\beta}$ follows a normal distribution with mean β and variance $\sigma_{\hat{\beta}}^2$:

$$\hat{\beta} \sim N \left[\beta, \sigma_{\hat{\beta}}^2 \right].$$

Standardize $\hat{\beta}$ so that it has a standard normal distribution:

$$\frac{\hat{\beta} - \beta}{\sigma_{\hat{\beta}}} \sim N [0, 1].$$

Suppose that we are interested in the null hypothesis

$$H_0 : \beta = \beta^0$$

against the alternative

$$H_1 : \beta \neq \beta^0.$$

Then, under the null hypothesis, we have

$$\frac{\hat{\beta} - \beta^0}{\sigma_{\hat{\beta}}} \sim N [0, 1].$$

$\sigma_{\hat{\beta}}$ is an unobserved population parameter. We can replace it with its unbiased estimator $s_{\hat{\beta}}$. This gives

$$\frac{\hat{\beta} - \beta^0}{s_{\hat{\beta}}} \sim t [\nu],$$

where ν denotes the degrees of freedom of the t distribution and is given by $n - K$. Then, we can state that

$$\text{Prob} \left(-t_{\alpha/2, \nu} < \frac{\hat{\beta} - \beta^0}{s_{\hat{\beta}}} < t_{\alpha/2, \nu} \right) = 1 - \alpha.$$

α is some probability value. $-t_{\alpha/2}$ and $t_{\alpha/2}$ are some lower and upper thresholds. These thresholds are typically referred to as the “critical values”. They correspond to points on the

x-axis of the t distribution. The interpretation of this expression is for the random variable $\frac{\hat{\beta} - \beta^0}{s_{\hat{\beta}}}$. The probability that this random variable is between the stated thresholds is $1 - \alpha$.

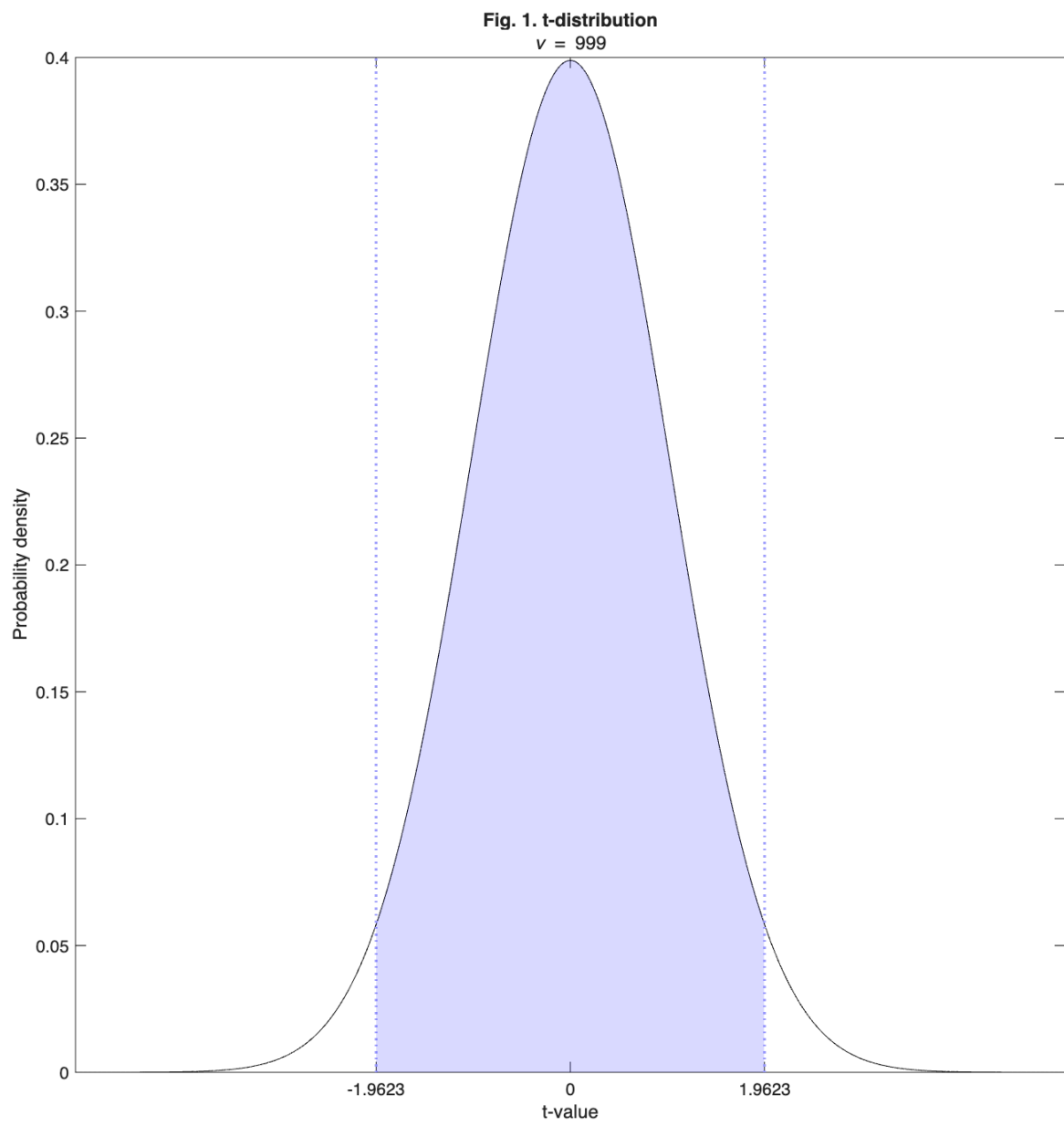
For example, if $\nu = 999$ and $\alpha = 0.05$,

$$t_{0.025} = 1.9623,$$

and hence

$$\text{Prob} \left(-1.9623 < \frac{\hat{\beta} - \beta^0}{s_{\hat{\beta}}} < 1.9623 \right) = 0.95.$$

Figure 1 illustrates this. It plots the PDF of the t distribution with 999 degrees of freedom. The shaded area between the stated thresholds is 95%.



Rearranging the terms of the probability expression, we obtain

$$\text{Prob} \left(\hat{\beta} - t_{\alpha/2} s_{\hat{\beta}} < \beta^0 < \hat{\beta} + t_{\alpha/2} s_{\hat{\beta}} \right) = 1 - \alpha.$$

Replace the null value β^0 with β because the confidence interval we are deriving is not intended to cover the null value but the true, fixed, population parameter. Therefore, we write

$$\text{Pr} \left(\hat{\beta} - t_{\alpha/2} s_{\hat{\beta}} < \beta < \hat{\beta} + t_{\alpha/2} s_{\hat{\beta}} \right) = 1 - \alpha.$$

The symbol β^0 plays a technical role in deriving the distribution of the t -statistic, whereas β denotes the unique, nonrandom population parameter that the confidence interval aims to cover in repeated sampling.

At this instance the interpretation changes. The interpretation is now for the unique non-random population parameter β . The end points of the interval,

$$\left[\hat{\beta} - t_{\alpha/2} s_{\hat{\beta}}, \hat{\beta} + t_{\alpha/2} s_{\hat{\beta}} \right],$$

are random. The interval does not take one value but a different value from one sample to another. Suppose that we randomly collect an infinite number of samples (repeated sampling, or sampling in the long run), and construct an interval for each sample. The stated probability tells that 95% of these intervals will include β . This is where the probabilistic interpretation comes from. Given the single sample data at hand, we could calculate only one interval estimate. Once we construct this interval using the sample data at hand, the end points of the interval are not random anymore. Therefore, the probability that β is in this interval is either 0 or 1. Hence, it is incorrect to say that the probability that β is in the interval we estimated is 95%. The interval we estimated (using the single sample at hand) is just an estimate of one of those intervals (that result from repeated sampling) that contain β 95 percent of the times. The correct interpretation is as follows. In repeated sampling, the probability that intervals like the one we estimated will contain β is 95%. The probability that this particular, non-random interval includes β is either 0 or 1.

The CI is also called the “interval estimate” because it provides a range of the possible estimates of the population coefficient, whereas, for example, the OLS estimate is a point estimate of the population coefficient. The CI can be seen as a possible measure of the precision of the point estimate. That is, once we obtain a point estimate, for example $\hat{\beta}$, we ask how precise we expect this estimate to be.

A test and a CI are closely related. We reject a null of the t test that $\beta^0 = 0$ because it lies outside the CI we calculate that does not include 0.

3. Application

3.1. Clear the memory

Clear the memory from possible calculations from an earlier session.

```
11 % 3.1. Clear the memory
12 clear;
```

3.2. Set the sample size

We assume that there are `N_obs` observations available for this independent variable and for the dependent variable of the regression. In our simulation exercise below, we will draw samples from the population. Setting the number of observations to `N_obs` means that we keep the sample size fixed at `N_obs` each time we draw a sample from the population.

```
14 % 3.2. Set the sample size
15 N_obs = 1000;
```

3.3. Generate data for the independent variable

We assume that the regression model contains an independent variable and, for simplicity, no constant term. The code presented in this section draws `N_obs` theoretical observations from the uniform distribution to create an artificial dataset for the independent variable X . It is not important which distribution we use. In our simulation exercise below, we will use these observations to generate data for y repeatedly. In the exercise we will keep the observations of X fixed. That is, as we will mimic taking repeated samples from the population, we will be doing this only for y and not for X . This means that we will keep the random data of X fixed in repeated sampling. Keeping X fixed in repeated sampling is indeed an assumption we make. Note, however, that this is the classical assumption we make while we derive the basic econometric theory. That is, we condition on the values of a regressor while we make econometric derivations. We do this because it simplifies the derivations, and the basics of econometric theory does not change.

```
17 % 3.3. Generate data for the only independent variable
18 X = random('Uniform',-1,1,[N_obs 1]);
```

3.4. Define the number of coefficients to be estimated

Define the number of coefficients to be estimated and assign it to the scalar array `N_par`. We will use `N_par` in few occasions in our code below.

```
20 % 3.4. Define the number of coefficients to be estimated
21 N_par = size(X,2);
```

3.5. Set a (hypothetical) value for the population coefficient

Assume that the population coefficient of the only independent variable β , that is `B_true` in the code, is equal to 0.5. We need this to generate values for y . In principle we do not observe β .

```
23 % 3.5. Set a (hypothesized) value for the population coefficient
24 B_true = 0.5;
```

3.6. Set the number of simulations

In this exercise a simulation refers to taking a random sample from the population. Since we want to take samples from the population repeatedly, we will be repeating the simulation multiple times. Here we define the number of simulations or samples.

```

26 % 3.6. Set the number of simulations
27 N_sim = 300;

```

3.7. Preallocate a matrix for storing OLS estimates from all samples

Create an empty matrix that will store the OLS coefficient estimates of β . Since we will draw `N_sim` samples from the population, we will obtain `N_sim` coefficient estimates based on these samples. Since we have only one coefficient to estimate, that is since `N_par` is 1, the matrix in this case is in fact a column vector.

```

29 % 3.7. Preallocate a matrix for storing OLS estimates from all samples
30 B_hat_sim = NaN(N_sim,N_par);

```

3.8. Preallocate a matrix for storing SSEs from repeated samples

Create an empty column vector that will store the `N_sim` standard error estimates of the OLS coefficient estimates of β from repeated samples.

```

32 % 3.8. Preallocate a matrix for storing SSEs from repeated samples
33 B_hat_SEE_sim = NaN(N_sim,N_par);

```

3.9. Coefficient and standard error estimates (SEE) from repeated samples

Here we draw `N_sim` random samples from the population as if we could do this in reality. Each sample leads to an estimate of β . This leads to a distribution of the OLS estimate, referred to as the sampling distribution of the OLS estimate of β – see the exercise on the sampling distribution of the OLS estimate.

We use a for loop to mimic drawing random samples from the population, and estimate the population coefficient of interest using each sample. In the code below, line 36 is the index of the for loop that instructs the for loop to execute the program we are to specify below `N_sim` times. In line 37, we draw random numbers for the error of the regression assuming that the errors follow a standard normal distribution. We specify the dimension of the errors as `N_obs` by 1. In line 38, we generate data for the dependent variable using the assumed true value for the population coefficient, the generated data for X , and the generated data for the error term. At each iteration of the for loop a new dataset is created for the dependent variable. In line 39, we estimate the regression equation using the data generated for y and X . For this purpose, we utilize an external function that takes a dependent variable and a matrix of independent variables as input arguments, and returns standard OLS statistics as output. In line 40, we store the coefficient estimate from iteration i of the for loop in row number i of the vector array `B_hat_sim`. Line 41 does this for the standard error estimate of the coefficient estimate. The last line closes the for loop.

Note that if we had samples of y and X at our disposal from repeated samples from the population, we would not need any of the code up to this point and instead we could start the exercise directly with the next section.

```

35 % 3.9. Coefficient and standard error estimates (SEE)
36 for i = 1:N_sim

```

```

37     u = random('Normal',0,1,[N_obs 1]);
38     y = X*B_true+u; % The data generating process (DGP)
39     LSS = lss(y,X);
40     B_hat_sim(i,1) = LSS.B_hat(1,1); % B_hat is a random variable
41     B_hat_SEE_sim(i,1) = LSS.B_hat_SEE(1,1);
42 end

```

4. Construct random intervals (RIs) from repeated samples from the population

We want to construct RIs. This requires to specify a significance level. Since we want to construct a 95% CI, we take this level as 5%. Calculate the degrees of freedom given the number of observations and parameters to estimate. Using these, calculate the critical value from the t distribution. Next, construct the intervals. Note that the dimension of the RIs array is N_{sim} by 2. That is, there are N_{sim} intervals resulting from N_{sim} samples. 2 is for the lower and upper bounds of the intervals. In a real life scenario, however, we typically have only one sample and hence we can estimate only one CI. In line 60 we extract such a CI from RIs we estimated.

```

44 %% 4. Construct RIs from repeated samples from the population
45
46 % 4.1. Define the significance level
47 alpha = 0.05; % For 95% CI. Change to 0.10 for 90% CI.
48
49 % 4.2. Calculate the degrees of freedom for the t distribution
50 nu = N_obs-N_par;
51
52 % 4.3. Calculate the critical value from the t distribution
53 t_c = tinv(1-alpha/2,nu);
54
55 % 4.4. Construct the RIs for B_true using its estimates from all
56 % samples
57 RIs = [B_hat_sim-t_c*B_hat_SEE_sim,B_hat_sim+t_c*B_hat_SEE_sim];
58
59 % 4.5. Construct the CI for B_true when there is one sample available
60 CI = RIs(1,:);

```

5. Plot the RIs from all samples and the population coefficient

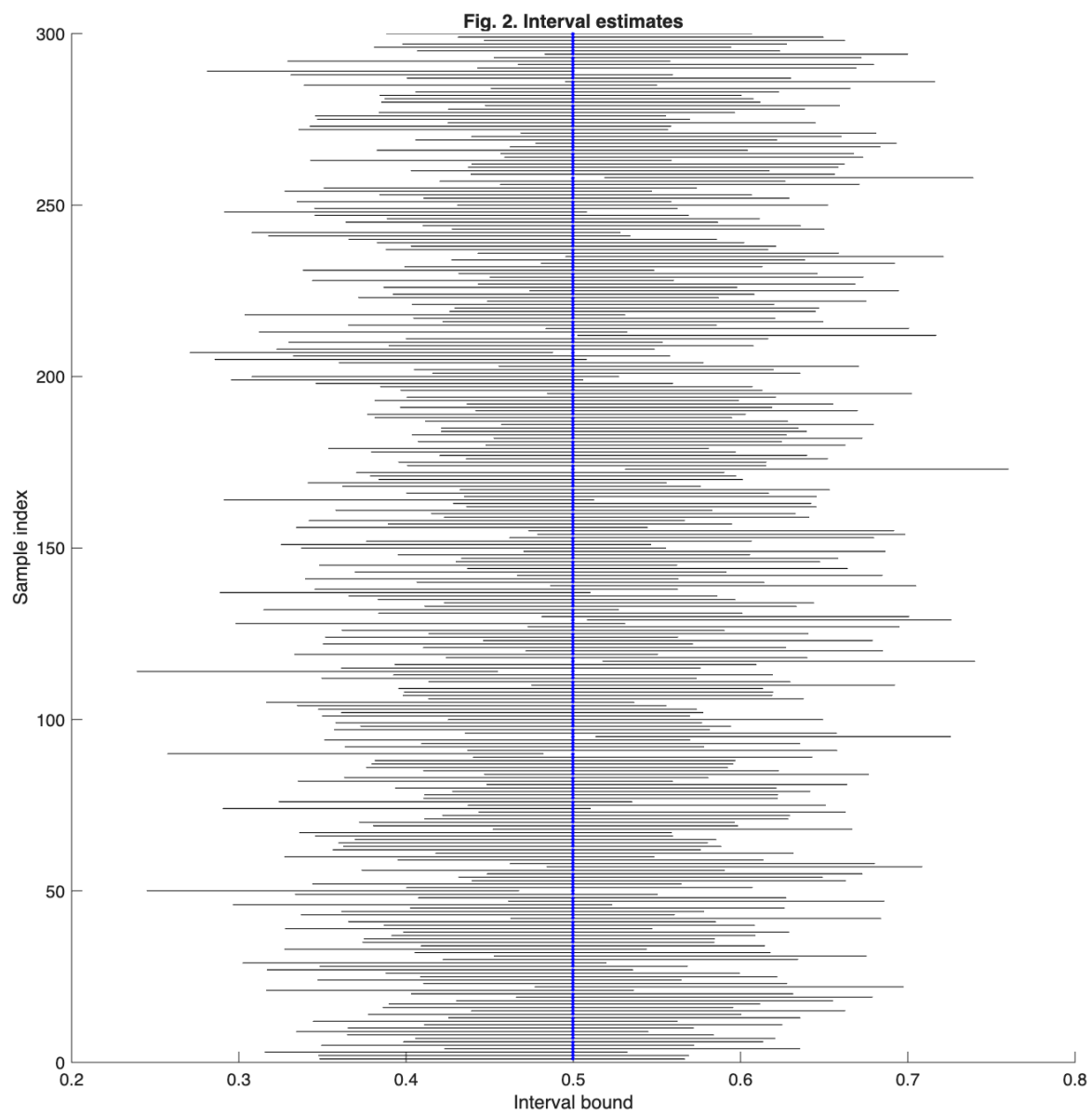
Here we plot the RIs we have estimated using the samples we have drawn from the population. We also plot the population coefficient.

```

61 %% 5. Plot the RIs from all samples and the population coefficient
62
63 % 5.1. Plot the RIs from all samples and the population coefficient
64 figure('Position',[100 100 1000 1000]);
65 hold on
66 for i = 1:N_sim
67     plot(RIs(i,:),[i,i],'k-','LineWidth',0.5);
68     plot(B_true,i,'bo','MarkerSize',1.5,'MarkerFaceColor','b');

```

```
69 end
70 title('Fig. 2.Interval estimates');
71 ylabel('Sample index');
72 xlabel('Interval bound');
73 hold off
```



6. Interpret the CI

Figure 2 shows all the RIs we estimated using repeated sampling, and the population coefficient. We can calculate the proportion of the times the population coefficient falls into these intervals. In line 77 we create a dummy variable that takes a value of 1 if the population coefficient falls within the intervals. In line 80 we calculate the proportion of the times the population coefficient falls into the intervals constructed using the repeated samples. The proportion we obtain is approximately 95%. It is not exactly 95% due to simulation noise.

We can now interpret the CI. In repeated sampling, the probability that intervals, like the one estimated using only one sample in line 60, contains the population coefficient β is 95%. The particular CI we have is called “confidence” interval because we use this, and only one interval to be confident about the population coefficient to a certain probability extent.

```
74 %% 6. Interpret the CI
75
76 % 6.1. Create a dummy indicating if B_true is within the RIs
77 B_true_is_within_RIs = B_true > RIs(:,1) & B_true < RIs(:,2);
78
79 % 6.2. Calculate the proportion of times B_true is within the RIs
80 Proportion_B_true_is_within_RIs = mean(B_true_is_within_RIs); % App.
81 % 0.95
```

7. Final notes

This file is prepared and copyrighted by Renata-Maria Istrătescu and Tunga Kantarcı. Parts of the theory section are based on Magnus, J. R., 2017. Introduction to the Theory of Econometrics. Amsterdam: VU University Press. This file and the accompanying MATLAB files are available on GitHub and can be accessed via this [link](#).